

Graph Diffusion Evolution Model for Multi-Conditional Molecular Generation

Xingcheng Fu
Key Lab of Education
Blockchain and Intelligent
Technology, Guangxi
Normal University
Guilin, China
Guangxi Key Lab of
Multi-source Information
Mining & Security,
Guangxi Normal University
Guilin, China
fuxc@gxnu.edu.cn

Lingyun Liu
Key Lab of Education
Blockchain and Intelligent
Technology, Guangxi
Normal University
Guilin, China
Guangxi Key Lab of
Multi-source Information
Mining & Security,
Guangxi Normal University
Guilin, China
liuyl@stu.gxnu.edu.cn

Yisen Gao
Computer Science and
Engineering, Hong Kong
University of Science and
Technology
Hong Kong, China
ygaodi@cse.ust.hk

Tianyu Chen
SKLCCSE, School of
Computer Science and
Engineering, Beihang
University
Beijing, China
tianyuc@buaa.edu.cn

Qingyun Sun*
School of Computer
Science and Engineering,
Beihang University
Beijing, China
Guangxi Key Lab of
Multi-source Information
Mining & Security,
Guangxi Normal University
Guilin, China
sunqy@buaa.edu.cn

Jianxin Li
SKLCCSE, School of
Computer Science and
Engineering, Beihang
University
Beijing, China
lijx@buaa.edu.cn

Xianxian Li
Key Lab of Education
Blockchain and Intelligent
Technology, Guangxi
Normal University
Guilin, China
lixix@gxnu.edu.cn

Abstract

The diffusion model with multiple conditions has received widespread attention in the field of drug design due to its high-quality generation ability. However, the paradigm of directly generating new molecules from conditions used in existing work has not accurately fitted the joint distribution of multiple conditions during the generation process. To address this issue, we propose **Graph Diffusion Evolution Model (GDEM)** for multi conditional molecule generation. GDEM decomposes the process of molecular generation into a chain-like Markov evolution process, continuously adjusting the molecular structure and gradually approaching the true multi-conditional joint distribution. Meanwhile, in order to effectively train this chain evolution generative model, we also propose a two-stage training approximation method to complete the training of intermediate steps. We validated the effectiveness of GDEM on multiple polymer datasets and small molecule datasets, and the results showed that GDEM has advantages in molecular properties and condition control compared to traditional methods.

*Corresponding author.



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CCS Concepts

• **Computing methodologies** → **Artificial intelligence; Machine learning**; • **Applied computing** → *Life and medical sciences*;

Keywords

Graph generation, Molecular generation, Diffusion model, Markov process

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1 Introduction

From a geometric viewpoint, many real-world systems—ranging from social and communication infrastructures to chemical interactions—can be described as large, intricate networks. Within this broader web of interconnected entities, molecules can be naturally modeled as graphs, making graph-based molecular generation particularly appealing: such models capture the distribution of molecular structures and efficiently constrain the chemical search space, thus supporting drug discovery and materials design [33, 34]. However, as practical applications often involve complex and interdependent requirements, generating molecules under a single

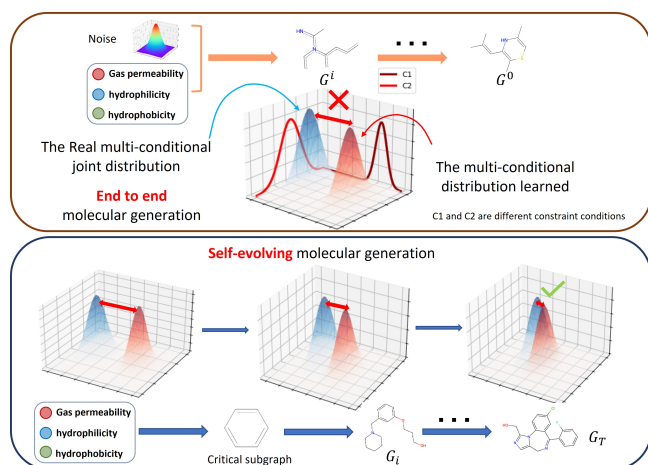


Figure 1: Traditional multiple conditionals molecular generation (top) is prone to bias when capturing conditional distributions, while the self-evolution process (bottom) can continuously fit the true joint distribution during the generation of molecules.

condition is no longer sufficient; it is increasingly crucial to develop approaches that can simultaneously satisfy multiple design constraints.

The existing methods combine multiple conditions into composite conditions, treating the generation of multiple conditions molecules as single composite condition generation and using it as a guide to achieve end-to-end generation [9, 20, 35]. However, the requirements of multiple conditions on molecular structure often conflict, and directly using composite condition for end-to-end generation can easily make it difficult for the model to converge or generate molecules that simultaneously meet multiple conditions [8, 29]. As shown in Figure 1, the distribution to which this composite condition conforms is offset or not fine enough compared to the real joint distribution of multiple conditions, which leads to the final generated molecules not meeting the constraints of multiple conditions well and not meeting practical needs.

Inspired by Markov process [3, 12], we try to design a graph markov evolution mechanism, which model the entire generation process as a Markov chain. This process is divided into three stages: (i) Founder Population: Just as chemists use functional groups to study molecules, a critical subgraph that simultaneously meets multiple conditions is first constructed as the cornerstone of evolution. (ii) Iterative Progeny Generation: Use the molecules of the previous generation as conditions to generate the next generation of molecules by graph diffusion model. (iii) Elite Selection: Select the molecule that best meets the target conditions as the condition for the next generation. This evolutionary generation method re-interprets the multi-condition molecular generation problem from the perspective of molecular structure, and continuously optimizes the molecular graph during the generation process, thereby enhancing the model’s ability to control conditions. However, this method still faces the following challenges: (1) *How to define or extract critical subgraphs that meet multiple conditions as the cornerstone of*

evolution? (2) *How can we train this evolutionary generation process efficiently, given that the discrete selection mechanism prevents gradient-based optimization?*

To address the above challenges, we propose Graph Diffusion Evolution Model (GDEM). Specifically, to address Challenge (1), inspired by invariant learning [15, 25, 31], we define the critical subgraph as a substructure in the original molecular graph (i.e., training set molecules) that invariant satisfies multiple conditions, and capture it from the original molecular graph through node importance encoding. Then, we designed a conditional control network to encode critical subgraphs and use them as conditions. Finally, we use the new molecules generated in the previous generation as conditions for the next generation generation, thereby continuously fitting the joint distribution of multiple conditions during the generation process. To address Challenge (2), we decompose the training process into a two-stage learning process: In the first stage, we learn to use critical subgraphs as conditions to generate new molecules, thus roughly capturing the joint distribution of multiple conditions. In the second stage, we directly sample new molecule from the distribution learned in the first stage as a new condition to learn the distribution of all intermediate generations. In the following text, we derive the upper bound of the error to prove that this is feasible. This two-stage fitting strategy effectively addresses the issue of gradient non-propagation caused by discrete selection within the error range.

Several experiments have shown the superiority of GDEM over baseline models. In summary, our contributions are as follows:

- **Novel Perspectives.** We regard molecular generation as a Markov evolutionary process, which continuously iterates and evolves based on critical subgraphs to approach the target conditional distribution.
- **Evolutionary Model.** We propose a two-stage approximate training method and design a graph-conditioned denoising network for effective training.
- **Better Performance.** Extensive experiments have demonstrated the significant performance improvement of our method in multiple conditions molecular generation.

2 Related Works

2.1 Graph Diffusion

The graph diffusion model has shown great potential in tasks such as molecular design and social network generation [21, 23, 43]. Among them, continuous latent space diffusion [2, 7, 24] maps the graph to a continuous vector for noise disturbance and denoising, which can directly draw on the continuous diffusion theory and efficient numerical tools in mature fields such as images. However, the discrete node and edge structures of the graph itself are prone to semantic distortion in continuous representation. The discrete diffusion method [39, 41] directly applies the diffusion process on the structure of the graph, strictly ensuring the structural validity of the generated results, but its computational cost is relatively high, requiring careful design of discrete state transition rules.

2.2 Molecular Generation

The core goal of molecular generation is to explore new functional molecules from the chemical space [5, 26]. Among them, the unconditional generation model [18, 30, 38] focuses on learning the basic laws of molecular data distribution, but its lack of directional control ability makes it unable to generate molecules that meet specific functional requirements. The subsequent single condition methods [10, 11, 13] can only respond to single attribute requirements, and when faced with multi-objective conflicts, the optimization process is prone to getting stuck in local optima. Finally, although multi-conditional molecular generation [22, 42] solves the above problems, existing work lacks precise capture of multiple conditional joint distributions, resulting in generated molecules that do not satisfy multiple conditional constraints.

3 Notation and Problem Definition

A molecular graph $G = (V, E)$ consists of a set of nodes (atoms) V and edges (bonds) E . Each atom is encoded as a one-hot vector $X_V \in R^{N \times F_V}$, where F_V denotes the number of atom types and $N = |V|$ is the number of atoms. Similarly, bonds are represented by a tensor $X_E \in R^{N \times N \times F_E}$, where F_E indicates the number of bond types (e.g., single, double, triple, aromatic).

Our model is a multi-conditional molecular generation model, whose objective is to generate molecules G that meet the condition set $C = \{c_1, \dots, c_M\}$, such as desired physicochemical properties, substructural motifs, or pharmacophoric constraints, where M is the number of properties. In other words, the goal of our task is to learn the joint conditional probability distribution of multiple conditions $p(G|C) = p(G|c_1, \dots, c_M)$.

In this paper, in order to consider both nodes and edges in both diffusion and generation processes, we use a single matrix $G \in R^{N \times F_G}$ to represent graph, with $F_G = F_V + N \cdot F_E$ [22].

4 Method

In this section, we will gradually introduce the detailed design of GDEM, its overall framework is shown in Figure 2. We first introduced our overall molecular evolution mechanism, followed by our critical subgraph extraction mechanism, and finally, we presented the graph diffusion evolution model.

4.1 Markov Evolution Generation

Our graph markov evolution mechanism consists of three parts:

(i) Founder Population: The differences in the requirements for molecular structures under various conditions result in the rarity of molecular structures that simultaneously meet multiple conditions. Therefore, we employ invariant learning to directly select a key substructure from the original molecular graph that simultaneously meet multiple conditions as the cornerstone of evolution, thus providing a specific and stable anchor point for the generation process.

(ii) Iterative Progeny Generation: Subsequently, we designed a graph-conditioned network similar to ControlNet [44], which can encode and inject graph conditions during the generation process. Then, we take the molecules generated in the previous iteration as the conditions for the current generation and continue the iteration. This method allows the conditional distribution of molecules to

change continuously during the generation process, rather than being fixed throughout end-to-end generation.

(iii) Elite Selection: In order to make the conditional distribution of molecules in the generation process gradually approach the joint distribution of multiple conditions in the correct way, we sample multiple new molecules at each generation, and then evaluate these molecules that best fits the joint distribution of multiple conditions by predicting their properties, and use this molecule as the generation result of this generation, as well as the new graph condition for the next generation. This elite selection strategy can ensure that the generation process approaches the joint distribution of multiple conditions from the correct path.

Specifically, the evolution process in the k -th generation can be defined as:

$$p(G|C) \approx \underbrace{\prod_{i=1}^k p(G_{i+1}|G_i, C)}_{\text{Graph Evolution}} * \underbrace{p(G_{sub}|C)}_{\text{Subgraph Extraction}}, \quad (1)$$

where the G_i is the optimal molecule selected at each generation. It should be noted that throughout the text, we use G^t to represent the final generated molecular graph in denoising timestep t , and use G_i to represent the conditional graph, which is a generated result. The G_{sub} in the paper can be regarded as G_0 .

4.2 Invariant Subgraph Extraction:

To identify critical substructures, we adopt an invariant learning approach, aiming to extract a subgraph $G_{sub} \subseteq G$ that consistently contributes to satisfying multiple conditions. To this end, we learn an extractor $f_\psi(G, C) \rightarrow G_{sub}$, which infers the subgraph G_{sub} from the input molecular graph G and a set of conditions C . Specifically, $f_\psi(\cdot)$ is implemented as a learnable node scoring module that assigns importance scores to nodes, enabling the decomposition of the molecular graph into causal-invariant and non-causal components. The causal-invariant part corresponds to the critical subgraph of interest.

In order to ensure that the captured critical subgraph G_{sub} conforms to multiple conditional constraints, we designed a multi-label predictor $f_\phi(\cdot)$. Our invariant subgraph loss is as follows:

$$\mathcal{L}_{inv} = \|f_\phi(f_\psi(G)) - C\|_2^2. \quad (2)$$

Due to significant differences between numerical and categorical conditions, we performed different loss functions for each domain. Specifically, for numerical conditions, we used Mean Squared Error(MSE) as the loss function, while for categorical conditions, we used Cross Entropy(CE) as the loss function. Finally, we linearly fused them into the final loss.

Additionally, we recommend adding sparsity constraints to ensure the accuracy of critical subgraphs:

$$\mathcal{L}_{spar} = \|\text{Mask}(G)\|_1. \quad (3)$$

Therefore, the full objective is as follows:

$$\mathcal{L}_{sub} = \mathcal{L}_{inv} + \lambda_{spar} \mathcal{L}_{spar}. \quad (4)$$

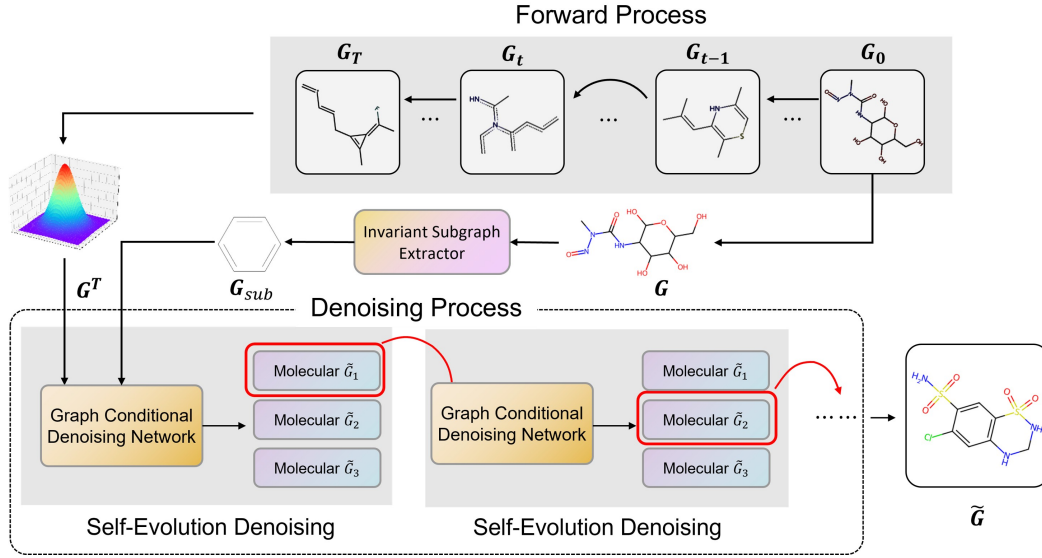


Figure 2: The overall framework of GDEM

4.3 Self-Evolving Diffusion Model

4.3.1 Forward diffusion process: In order to better capture the relationship between nodes and edges, we adopted a forward diffusion process that associates nodes and edges [22]. Specifically, the transition matrix Q_G considers the joint distribution of nodes and edges. Formally, Q_G can be represented as:

$$Q_G = \begin{bmatrix} Q_V & \mathbf{1}'_N \otimes Q_{VE} \\ \mathbf{1}_N \otimes Q_{EV} & \mathbf{1}_{N \times N} \otimes Q_E \end{bmatrix}, \quad (5)$$

where $\otimes$ represents Kronecker product, $\mathbf{1}_N$, $\mathbf{1}'_N$, and $\mathbf{1}_{N \times N}$ represent column vectors, row vectors, and matrices of all 1 element. Here, we choose the transition matrices $Q_{VE} \in R^{F_V \times F_E}$ and $Q_{EV} \in R^{F_E \times F_V}$ by calculating the co-occurrence frequency of atoms and bond types in the training molecule graph to derive the marginal atomic-bond co-occurrence probability distribution. Therefore, the forward diffusion process can be defined as:

$$q(G^t | G^{t-1}) = \text{Cat}(G^t; G^{t-1} Q_G^t), \quad (6)$$

where $t \in [1, T]$ denotes the timestep of adding noise.

4.3.2 Reverse process: Considering that multi-condition generation requires the simultaneous determination of the properties of multiple different molecules, we adopt classifier-free conditional generation method to avoid additional computational consumption. Specifically, it can be denoted as:

$$\begin{aligned} \hat{p}_\theta(G^{t-1} | G^t, G_i, C) &= s \log p_\theta(G^{t-1} | G^t, G_i, C) \\ &+ (1-s) \log p_\theta(G^{t-1} | G^t, \emptyset), \end{aligned} \quad (7)$$

where s is the hyperparameter to control the condition guidance.

Specifically, in the generation stage, guided by the original condition C , the graph condition G_i , we iteratively denoise step by step, starting from the noise, which can be expressed as:

$$G^{t-1} \sim q(G^{t-1} | \hat{G}^0, G^t) p_\theta(\hat{G}^0 | G^t, G_i, C), \quad (8)$$

where θ denotes the denoising model. It is worth noting that since only C is available during the generation process, and the conditional graph G_i is unknown, we need to select a molecular graph G with similar conditions from the training set based on C to obtain G_{sub} as the initial conditional graph.

4.3.3 Elite selection: During the iterative generation process, we sample M molecules at each generation i and select the one that best meets condition C as the Conditional graph G_i for the next reverse generation process. By repeating the above process, the GDEM can self-adjust the distribution of the conditional graph during the generation process, so that the generated molecules can continuously evolve and approach the true multiple conditional joint distribution. This is the key to our proposed self-evolution process.

4.3.4 Two-stage training: The self-evolving denoising model $f_\theta(\cdot)$ needs to be trained under different graph conditions to fit to the distribution of intermediate steps in the generation process. However, the method of multiple sampling and selecting the optimal result used in the generation process disrupts the propagation of gradients, making it impossible for us to train each generation simultaneously. Therefore, we propose a two-stage training approximation method to complete the training of intermediate steps.

In the first stage, we only train model $f_\theta(\cdot)$ on how to generate molecular graphs under the guidance of critical subgraph G_{sub} . The loss function $\mathcal{L}_1$ of this process can be represented as:

$$\mathcal{L}_1 = \|f_\theta(G^t, G_{sub}, C) - G\|_2^2. \quad (9)$$

In the second stage, we believe that the G_1 , sampled from the first stage is already very close to the multiple conditions joint distribution, so we use G_1 as the condition to approximate each subsequent training step. Specifically, we only need to replace the

conditional graph with G_1 to retrain the model:

$$\mathcal{L}_2 = \|f_\theta(G^t, G_1, C) - G\|_2^2. \quad (10)$$

We further derived the error of this approximation method, theoretically ensuring its feasibility.

THEOREM 4.1. *Assume that each step in the diffusion chain introduces a bounded approximation error. That is, for all i ,*

$$D(p(G_{i+2} | G_{i+1}, C) \| p(G_{i+1} | G_i, C)) \leq \epsilon_i. \quad (11)$$

Also, assume that the initial subgraph G_1 , generated from condition y , approximates the full graph distribution with KL divergence bounded by $D(p(G | G_1, C) \| p(G | C)) \leq \delta$. To simplify the notation, we use $p(G_c)$ to denote $p(G | C)$. Then, when evaluated under the true target distribution $p(G_c)$, the additional error incurred by approximating $p(G_k | G_{k-1}, C)$ with $p(G_k | G_1, C)$ is bounded by:

$$D(p(G_c) \| p(G_k | G_1, C)) - D(p(G_c) \| p(G_k | G_{k-1}, C)) \leq \delta + \sum_{i=1}^{k-2} \epsilon_i, \quad (12)$$

where the δ is affected by the first stage and can be reduced through training in the first stage, while ϵ is a very small value that describes the difference between two intermediate distributions. Therefore, when k is not large, the error of this method can be accepted.

4.3.5 Graph conditional denoising network: We innovatively propose a graph conditional denoising network to guide molecular generation to adapt to our proposed self-evolving generation process. This denoising network consists of a backbone network, an initial condition encoder and a graph conditional control network inspired by ControlNet [44].

Firstly, in order to process data from different domains, we use an initial condition encoder [22]. Specifically, for category conditions, we encode them using a one-hot vector, and for numerical conditions, we encode them using clustering methods. In practice, we implement it as two *Linear* layers and a *Softmax* layer:

$$H_{con} = \text{Linear}(\text{Softmax}(\text{Linear}(C))). \quad (13)$$

Secondly, the backbone network $f_\mu(\cdot)$ of our denoising network is a graph transformer layer [22] composed of multi-head attention mechanism and normalization layer. Specifically, we first encode the noisy graph G^T :

$$H_G = \text{MHA}(\text{MLP}(G^T)), \quad (14)$$

the $\text{MHA}(\cdot)$ here is a multi-head self-attention module. Then, we feed the encoded results into the conditional normalization layer $\text{GDLN}(\cdot)$ to obtain conditional results. For each row h_G in H_G and h_{con} in H_{con} :

$$\text{GDLN}(h_G, h_{con}) = \alpha_\pi(h_{con}) \odot \frac{h_G - \mu(h_G)}{\sigma(h_G)} + \beta_\pi(h_{con}), \quad (15)$$

where the $\alpha_\pi(\cdot)$ and $\beta_\pi(\cdot)$ are neural network modules about mean and variance in $f_\mu(\cdot)$. Repeat the above operation once and replace MHA with MLP to obtain the final output Z_{out} .

Subsequently, inspired by ControlNet, we have designed a graph based conditional control network. We freeze and copy the backbone network $f_\mu(\cdot)$ to obtain $f'_\mu(\cdot)$, and then fuse the conditional graph with the input t -th step noise graph G^t of the backbone

network through a zero graph convolutional layer to maintain dimensionality and obtain the input Z of $f'_\mu(\cdot)$:

$$Z = f_{zero}(G_i) + G^t, \quad (16)$$

where the G_i is a conditional graph, which is generated as G_{sub} in the first generation, and the $f_{zero}(\cdot)$ is a zero convolutional layer.

Then, the fused Z is passed as a condition to the graph transformer layer and another $f_{zero}(\cdot)$, and then added to the output of $f'_\mu(\cdot)$ to maintain its dimensionality, resulting in a new molecule G' :

$$G' = f_{zero}(f'_\mu(Z, G^t)) + Z_{out}. \quad (17)$$

Finally, we obtained the result G' of one-step graph conditional denoising. After fusing the G' and the results of unconditional denoising, we can obtain the final molecule G^{t-1} generated in this step.

5 Experiment

In this section, we conducted extensive experiments on a large polymer dataset and three small molecule datasets to validate the quality of the molecules generated by GDEM. In addition, we also compared our results with existing generation methods on multiple metrics, and the results showed that our method achieves the best performance in distributed learning and conditional control. We will demonstrate the advantages of our method through the performance comparison section. We also conducted ablation experiments to demonstrate the effectiveness of our important module. Finally, we visualized the generated results.

5.1 Dataset Description.

We conducted experiments on four datasets, including: O₂-N₂-CO₂ [40], HIV, BBBP, and BACE [22, 36]. Among them, O₂-N₂-CO₂ is the polymer material dataset, which contains three gas permeability conditions: O₂Perm, N₂Perm, and CO₂Perm. In addition, HIV, BBBP, and BACE are datasets belonging to the field of drug design, which respectively contain conditions HIV virus replication inhibition, blood-brain barrier permeability, and human β -secretase 1 inhibition. In addition, we also added two conditions for synthetic accessibility scores (SAS) and complexity scores (SCS) to each dataset [4, 6, 22]. The statistical data of these four datasets are presented in Table 2.

5.2 Experimental Setup

Metrics. To validate the effectiveness of our proposed GDEM model, we randomly split the dataset into three subsets, with 60% for training, 20% for testing, and the final 20% for validation. We use metrics [27] (1) Fréchet ChemNet Distance with the reference set (Distance) [28], MAE between the generated and conditioned, (2) synthetic accessibility score (Synth.) [6], (3) ~ (5) MAE/Accuracy for the numerical/categorical task conditions (Property) to evaluate the performance of the model.

In the above metrics, (1) is used to evaluate the distribution gap between the molecules generated by the model and the reference molecules, that is, to evaluate the distribution learning ability of the model. (2) ~ (5) are used to evaluate the degree to which the generated molecules satisfy various constraint conditions, that is, to assess the model's ability in controlling conditions.

Tasks	Model	Distribution Learning Distance ↓	Synthe. MAE ↓	Condition Control Property Acc. ↑	Avg. Rank ↓
Synth. & BACE	Graph-GA	7.4104 ±0.042	0.9633 ±0.041	0.4690 ±0.085	7.0000
	MARS	<u>6.7923</u> ±0.031	1.0123 ±0.021	0.5184 ±0.084	5.5000
	LSTM-GB	17.5585 ±0.031	0.9207 ±0.254	0.5816 ±0.022	3.5000
	JTVAE-ET	30.4696 ±0.084	0.9923 ±0.089	0.4628 ±0.034	8.0000
	DiGress	24.6560 ±0.032	2.0681 ±0.043	0.5061 ±0.016	8.5000
	DiGress-v2	25.3270 ±0.023	2.3365 ±0.086	0.5113 ±0.03	8.0000
	GDSS	46.7539 ±0.223	1.6422 ±0.043	0.5036 ±0.012	8.0000
	MOOD	44.2394 ±0.021	1.8853 ±0.043	0.5062 ±0.014	7.5000
	Graph-DiT	7.0456 ±0.021	<u>0.4466</u> ±0.01	<u>0.9168</u> ±0.037	<u>1.5000</u>
	GDEM (Ours)	6.1629 ±0.030	0.4079 ±0.021	0.9205 ±0.021	1.0000
Synth. & BBBP	Graph-GA	10.1659 ±0.097	1.2082 ±0.031	0.3015 ±0.032	8.0000
	MARS	10.9791 ±0.103	1.2250 ±0.032	0.5189 ±0.022	6.5000
	LSTM-GB	16.3904 ±0.0132	0.9969 ±0.041	0.5590 ±0.045	4.5000
	JTVAE-ET	33.5746 ±0.021	1.1619 ±0.023	0.4958 ±0.032	6.5000
	DiGress	18.6921 ±0.213	2.3658 ±0.031	0.6536 ±0.022	7.0000
	DiGress-v2	19.4498 ±0.0123	2.2694 ±0.077	0.6531 ±0.022	7.0000
	GDSS	39.9440 ±0.210	1.3788 ±0.021	0.5037 ±0.031	7.5000
	MOOD	34.2506 ±0.114	2.0284 ±0.065	0.4903 ±0.017	9.0000
	Graph-DiT	11.8519 ±0.224	0.3522 ±0.032	<u>0.9413</u> ±0.045	<u>1.5000</u>
	GDEM (Ours)	<u>10.8433</u> ±0.014	<u>0.4557</u> ±0.012	0.9593 ±0.034	1.0000
Synth. & HIV	Graph-GA	4.4418 ±0.211	0.9839 ±0.058	0.6035 ±0.011	5.0000
	MARS	7.2893 ±0.085	0.9691 ±0.064	0.6455 ±0.021	4.0000
	LSTM-GB	7.4659 ±0.171	0.9480 ±0.078	0.6736 ±0.041	3.0000
	JTVAE-ET	41.9771 ±0.424	1.2359 ±0.067	0.4850 ±0.070	7.5000
	DiGress	13.0409 ±0.145	1.9216 ±0.094	0.5335 ±0.097	7.5000
	DiGress-v2	13.3997 ±0.0167	1.5934 ±0.021	0.5331 ±0.007	7.5000
	GDSS	45.3416 ±0.348	1.2515 ±0.067	0.4830 ±0.013	8.5000
	MOOD	32.3523 ±0.311	2.3144 ±0.097	0.5106 ±0.072	9.0000
	Graph-DiT	6.0061 ±0.111	0.3086 ±0.063	<u>0.9651</u> ±0.077	<u>1.5000</u>
	GDEM (Ours)	<u>5.9551</u> ±0.022	<u>0.3207</u> ±0.02	0.9806 ±0.021	1.0000

Table 1: The generation result table of three small molecule datasets (BACE, BBBP, and HIV), where Synth. is the synthesizability score (Bold: best; Underline: runner-up, ↓ indicates that a smaller value is better, while ↑ indicates that a larger value is better).

Datasets	# Molecule	Avg. # Atoms	Avg. # Bonds	# Conditions
O ₂ -N ₂ -CO ₂	553	27.97	32.67	5
BACE	1332	33.67	36.44	3
BBBP	872	24.38	26.26	3
HIV	2372	25.35	27.28	3

Table 2: All datasets used for multi conditional molecular tasks.

Implementation Details. The parameters of baselines are set to the default values in the original papers. For GDEM, we set a diffusion step size of 500, and when calculating the loss in the one-step denoising process, we assign a loss weight of 1:10 to the nodes and edges. In addition, for the critical subgraph extraction module, we define the proportion of subgraph nodes as 30% and preserve the connection of the nodes in the complete molecular graph. Finally, for the multi-sampling operation, in order to balance the computational cost and performance simultaneously, we set the sampling quantity to 3.

Hardware and Software Configurations. **Hardware and Software Configurations.** We conduct the experiments with Ubuntu 20.04.6 LTS operating system, Intel(R) Xeon(R) Gold 6330 CPU @ 2.00GHz CPU and A800 80G GPU, our running memory is 120G, In addition, our main software platforms are CUDA 11.8.0, Python 3.9.20, PyTorch 2.4.0 and PyTorch-Geometrics 2.6.1.

5.3 Baseline Approaches

To validate the effectiveness of our proposed GDEM model, we compare it with various baseline approaches from two categories:

5.3.1 Molecular optimization baselines: we compare our method against several representative molecular optimization baselines. Graph-GA [14] uses a graph-based genetic algorithm with Monte Carlo tree search for chemical space exploration. MARS [37] performs multi-objective molecular optimization via Markov chain sampling. JTVAE [16] generates molecular graphs by encoding tree-structured scaffolds to ensure chemical validity(JTVAE-ET). LSTM-based models(LSTM-GB), as evaluated in the Guacamol benchmark suite [1], models treat SMILES as sequences and generate molecules using recurrent neural networks, as benchmarked in Guacamol.

Model	Distribution Learning	Condition Control				
	Distance ↓	Synth. ↓	O ₂ ↓	N ₂ ↓	CO ₂ ↓	# Avg. MAE ↓
Graph-GA	9.1882 ±0.064	1.3307 ±0.071	1.9840 ±0.010	2.2900 ±0.021	1.9489 ±0.013	1.8884 ±0.011
MARS	7.5620 ±0.024	<u>1.1658</u> ±0.044	1.5761 ±0.023	1.8327 ±0.021	1.6074 ±0.034	1.5455 ±0.017
LSTM-GB	18.1562 ±0.055	1.4251 ±0.034	1.1003 ±0.094	1.2365 ±0.011	1.0772 ±0.023	1.2098 ±0.014
JTVAE-ET	23.5990 ±0.024	1.0714 ±0.034	1.0781 ±0.071	1.2352 ±0.041	1.0978 ±0.064	1.1206 ±0.078
DiGress	22.7237 ±0.044	2.9842 ±0.064	1.7163 ±0.097	2.0630 ±0.054	1.6738 ±0.084	2.1093 ±0.121
DiGress-v2	21.7311 ±0.151	2.7507 ±0.044	1.7130 ±0.099	2.0632 ±0.067	1.6648 ±0.087	2.0479 ±0.078
GDSS	34.2627 ±0.078	1.3701 ±0.047	1.0271 ±0.074	1.0820 ±0.024	1.0683 ±0.064	1.1369 ±0.097
MOOD	39.3981 ±0.057	1.4019 ±0.097	1.4961 ±0.091	1.7603 ±0.035	1.4748 ±0.083	1.5333 ±0.021
Graph-DiT	6.6473 ±0.083	1.3497 ±0.031	<u>0.7440</u> ±0.096	<u>0.8857</u> ±0.012	<u>0.7550</u> ±0.067	<u>0.9336</u> ±0.023
GDEM(Ours)	<u>6.9363</u> ±0.033	1.2856 ±0.021	0.7142 ±0.023	0.8379 ±0.021	0.7104 ±0.043	0.8870 ±0.009

Table 3: The generation result table of O₂-N₂-CO₂ polymer dataset, where Synth. is the synthesizability score (Bold: best; Underline: runner-up, ↓ indicates that a smaller value is better, while ↑ indicates that a larger value is better).

5.3.2 Diffusion baselines: We further compare with recent diffusion-based generative models. GDSS [17] models graph generation as a stochastic differential system to capture complex structural patterns. DiGress [32] applies discrete denoising diffusion tailored for molecular graph generation. MOOD [19] explores out-of-distribution molecule generation guided by learned score functions. DiGress-v2 [32] extends DiGress [32] with conditional generation capabilities for property-controlled molecule design. Graph-DiT [22] fuses multiple conditions by processing conditions from different domains differently.

5.4 Performance Comparison

To comprehensively evaluate the performance of our proposed Graph Diffusion Evolution Model (GDEM), we conduct experiments on both polymer and small molecule datasets, and compare it with a series of representative baselines. The results are presented in Table 1 and Table 3, and we analyze them from three critical perspectives: distribution learning, condition control, and overall performance. In addition, our evaluation results all used 10-fold cross-validation.

5.4.1 Distribution learning: Distribution learning assesses how well the generated molecular graphs match the real data distribution. This reflects the model’s ability to explore and generalize chemical space realistically.

Graph-GA [14] generates molecules within the distribution, while MARS [37] directly edits molecular segments, both of which have achieved good results in distribution learning. However, in comparison, our method is still optimal.

This superiority comes from GDEM’s two-stage diffusion training mechanism. In the first stage, it captures a coarse approximation of the joint distribution by generating molecules from critical subgraphs derived via invariant learning. In the second stage, it fine-tunes this approximation through a Markovian evolution process that learns the transition distributions between intermediate graph states. This staged approach ensures that GDEM does not collapse to mode-specific or distributionally narrow outputs, which is a common issue in single-stage or one-shot generation frameworks like JTVAE [16].

5.4.2 Condition controllability: Generating molecules that meet multiple property constraints simultaneously is essential in real-world applications such as drug design and materials discovery. GDEM is specifically designed to address this multi-conditional generation challenge.

We achieved the best performance on both the polymer dataset and the small molecule dataset, outperforming the baseline model in almost all condition control aspects.

These results demonstrate that GDEM can fine-tune the conditional distribution during generation in a way that previous models cannot. While prior methods (e.g., DiGress-v2, MOOD) impose conditions directly on initial noise or latent codes, GDEM incorporates subgraph-based condition encoding and a stepwise refinement process. This allows GDEM to correct deviations from the desired property constraints progressively, rather than relying on a static, one-shot condition alignment.

5.4.3 Overall Performance. Traditional baselines like Graph-GA [14] and LSTM-GB [1] may perform well on synthesizability or certain properties, but lack balance across all metrics. DiGress-based [32] methods, while diffusion-driven, suffer from either high distribution distance or poor property alignment due to their lack of structured intermediate control. GDEM’s self-evolution generation structure, which decomposes the generation into controllable sub-processes, achieves a better trade-off among synthesizability, property satisfaction, and distribution fidelity.

5.5 Ablation Study

We validated the effectiveness of two important mechanisms in GDEM through ablation experiments: critical subgraph conditions and self-evolving chain generation, as shown in Table 4 and Table 5, the MDG is the original model without critical subgraph conditional mechanism and self evolutionary generation mechanism, GDEM(w/o s) and GDEM(w/o c) are partial models without critical subgraph conditional mechanism and self evolutionary generation mechanism, respectively. The experimental results showed that after removing the two mechanisms we proposed, the conditional control and distribution learning abilities of the model significantly decreased, which strongly demonstrates the effectiveness of our proposed mechanisms.

Model	Distribution Learning	Condition Control				
	Distance ↓	Synth. ↓	O ₂ ↓	N ₂ ↓	CO ₂ ↓	# Avg. MAE ↓
MDG	9.7716 ±0.044	1.5798 ±0.024	1.2163 ±0.063	1.4916 ±0.087	1.1761 ±0.021	1.3660 ±0.089
GDEM(w/o subg)	8.6772 ±0.022	<u>1.3025</u> ±0.101	0.9873 ±0.014	<u>1.0301</u> ±0.072	0.9078 ±0.021	<u>1.0569</u> ±0.048
GDEM(w/o Cha)	7.3215 ±0.099	1.4354 ±0.021	0.9080 ±0.072	1.0833 ±0.034	0.9058 ±0.067	1.0831 ±0.055
GDEM	6.9363 ±0.114	1.2856 ±0.077	0.7142 ±0.067	0.8379 ±0.024	0.7104 ±0.034	0.8870 ±0.065

Table 4: The results of the ablation experiment on O₂-N₂-CO₂ polymer dataset (Bold: best; Underline: runner-up, ↓ indicates that a smaller value is better, while ↑ indicates that a larger value is better).

Model	Distribution Learning	Condition Control	
	Distance ↓	Synth. ↓	Property Acc. ↑
MDG	13.6889 ±0.244	1.0808 ±0.072	0.7818 ±0.024
GDEM(w/o s)	12.0153 ±0.024	0.6155 ±0.014	0.8732 ±0.034
GDEM(w/o c)	<u>11.3313</u> ±0.044	<u>0.5554</u> ±0.034	<u>0.9173</u> ±0.042
GDEM	10.8433 ±0.094	0.4557 ±0.014	0.9592 ±0.034

Table 5: The results of the ablation experiment on small molecule dataset BBBP (Bold: best; Underline: runner-up, ↓ indicates that a smaller value is better, while ↑ indicates that a larger value is better).

6 Limitation

Although GDEM achieves notable progress in multi-conditional molecular generation, several open challenges remain that we view as natural next steps rather than overlooked issues. The first lies in its reliance on the invariant subgraph extractor and the property predictor for elite selection: these auxiliary modules are necessary for enforcing conditional guidance, yet any bias they introduce can influence the downstream diffusion process. A promising direction is to integrate these components more tightly with the denoiser, for example through joint fine-tuning and stability-aware regularization, which we leave for future investigation. In addition, the self-evolving multi-sampling procedure improves conditional accuracy but leads to high computational cost, restricting scalability to large molecules or expensive property evaluators. Reducing this cost through diffusion-step distillation or adaptive sampling remains an important engineering goal.

7 Conclusion

This article proposes a Graph Diffusion Evolution Model (GDEM) for multi-conditional molecule generation, which reconstructs traditional end-to-end molecule generation into a chain generation from conditions to critical subgraphs and then to self-evolution of molecules, achieving more accurate fitting of real multi-conditional joint distributions and enabling the generated molecules to better meet multi-conditional constraints and practical needs. In order to effectively train this chain model, we also propose a two-stage training approximation method to achieve training for intermediate steps. Subsequently, we validated the superiority of GDEM by comparing it with the baseline model on a large polymer dataset and three small molecule datasets, and visualizing it under different multiple conditions constraints. Finally, we have identified some limitations of our work, including exploring potential biases in GDEM and considering computational efficiency, which will be the focus of our future work.

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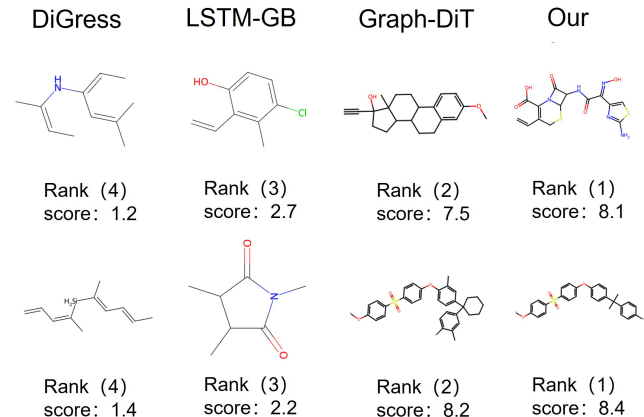


Figure 3: The comparative results of molecules generated by different models and our method are presented, with two rows using different condition.

5.6 Visualization

In this section, we present the molecular generation results of three baseline models and GDEM under two specific multiple conditions constraints: { SAS=2.4, SCS=3.51, P_{NP}=1 } (upper), { SAS=5.85, SCS=2.81, O₂Perm=0.98, N₂Perm=0.16, CO₂Perm=13.8 } (below). We rank based on conditional-error scores, as shown in Figure 3, baseline models often satisfy only part of the conditions or yield implausible scaffolds. In contrast, the GDEM we proposed can better meet all the specified constraints while retaining the chemically viable scaffold. These visual results corroborate the quantitative metrics and highlight GDEM’s strength in capturing multi-conditional molecular patterns.

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